

Measuring g Using a Bouncing Tennis Ball

Purpose

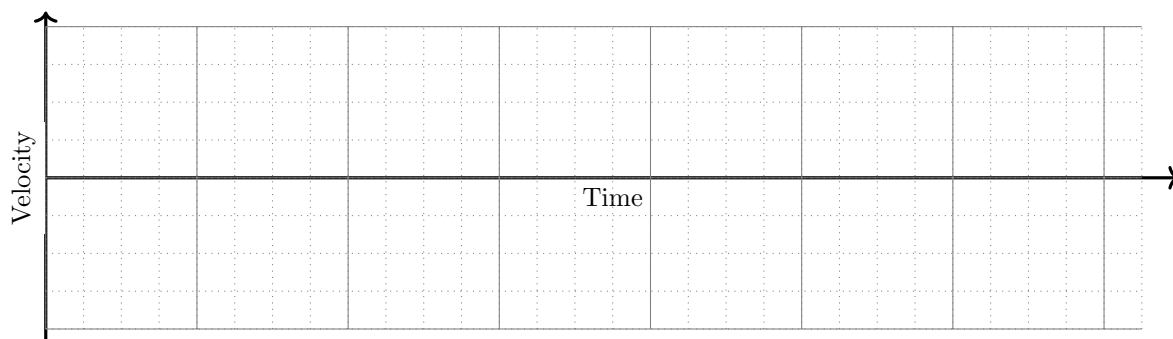
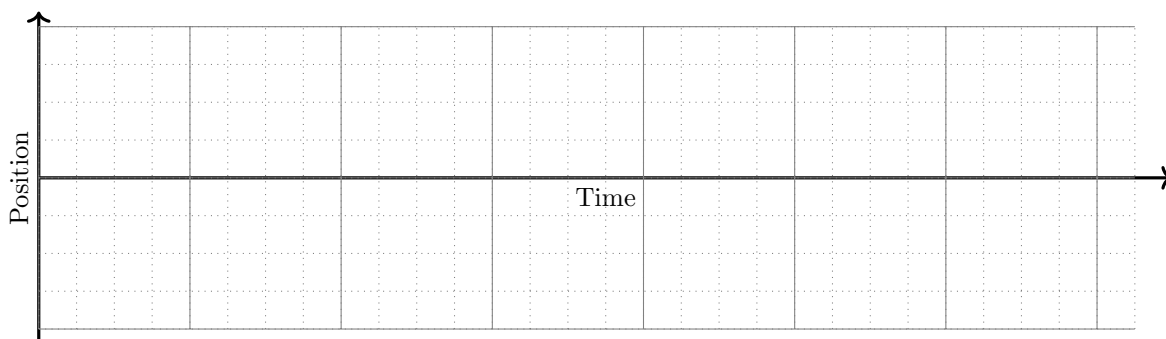
Use the graphs of motion for a ball that is in free fall to measure the acceleration of gravity g .

Procedure

1. Plug the LabQuest Mini into your computer
2. Download `Measuring-g-lab.gaml` from Schoology and open it in **Graphical Analysis**.
3. Hold the ball about 50 cm away from the motion detector. Start taking data, and then drop the ball, allowing it to bounce a few times.
4. See if your data looks smooth. If there have been issues, take a new data set before proceeding.

Data

Sketch the position and velocity graphs below. Please **be careful** with the horizontal axis. Try to make it so that each graph has the same events occurring at the same location along the t -axis. Don't rush this step.



Analysis

5. On each graph, label all “interesting” times. For example, label when the original drop time is, when the bounces occur, and when the maximum heights occur.
6. Explain why each graph makes sense given the motion of the ball.

- Position graph

- Velocity graph

- Acceleration graph

7. What does a linear segment of a velocity vs. time graph indicate? What is the significance of the slope of that linear segment?

8. The velocity graph should be linear in the region where the ball was in freefall. Fit a linear equation to your data in this region.

- Select the data in the region that corresponds to when the ball was in freefall.
- Click Graph Tools, , for the velocity graph and choose Apply Curve Fit.
- Select Linear as the curve fit and click Apply.

Write down the equation for the best fit line:

9. Use your best fit equation to identify the measured value of g .

10. Calculate percent error based on the expected value for g .

$$\text{Percent Error} = \frac{|\text{measured} - \text{expected}|}{\text{expected}} \times 100\%$$

Name:

Date:

Period:

Conclusion

Write a paragraph explaining your results on the lab. Your paragraph should address: (1) Explain what experiment you did and what graphs you made; (2) Explain what the graphs look like and why those shapes made sense; (3) Explain how you got your measured value for g and a report of the percent error compared to the expected value; and (4) Identify three to four factors that could have created uncertainties in your measurements and explain how each factor may have influenced your percent error.